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Sensai — Feature Catalogue

This document describes the available features for the Sensai project. For each feature, you will find a small description, real-world usage examples, and its rating.
The rating is designed to help you arbitrate your scope. (A group that chooses fewer well-integrated, well-justified features will be evaluated more favorably than a group that accumulates features without depth)

Descrptions are (by design) not very precise in order to let you find in your own words why those features are interesting and pitch them during your keynote. If you have doubt about whether what you're implementing counts as a certain feature, ask this question during the Follow-up.



AI is a fast-evolving domain. We've done our best to research features from the landscape, but if you see something missing, ask your pedago to submit it to us. Thank you for this collaborative effort!

🔧 Base — Mandatory for all groups

[B1] Functional CLI chatbot (👍)

The common starting point. A command-line chatbot connected to a local model via Ollama, with response streaming and conversation history management within a session.
Real-world examples: Claude CLI, LM Studio terminal interfaces, some local chat scripts you'll find on GitHub.

🧠 Memory & Context

[M1] Session Persistence & Profiles (⭐)

Save and reload the messages history across distinct sessions, plus a persistent user profile (preferences, custom instructions) auto-injected at init.

Real-world examples: ChatGPT's ”custom instructions”. A medical assistant that knows a user is allergic to penicillin without them mentioning it every time. A code assistant that knows you work in Python without you repeating it.

[M2] Token Budgeting & Semantic Compression (⭐⭐ + Depends on metrics)

Count token consumption per turn against a set budget. As you approach the cap, trigger a sliding-window summarization pass that condenses the oldest turns into a compact block, freeing space while preserving critical facts.

Real-world examples: Claude Projects summarizing older exchanges to maintain coherence over long working sessions. Customer support AI systems condensing a long ticket's history.

[M3] External Structured Memory (⭐⭐)

Move memory from flat files to a real database (SQLite/NoSQL). The agent reads, writes, and updates structured records — facts, entities, relationships — through CRUD operations it drives itself across long-term interactions.

Real-world examples: A CRM AI that updates a client record throughout interactions. A research assistant that maintains an enriched fact base across sessions. ”Second Brain” architectures (`coleam00/second-brain-starter`).

[M4] Artifact & State Document (⭐⭐)

The agent maintains a separate living document that it updates incrementally across turns, distinct from the raw chat history. The document is a deliverable, not a transcript. The model must update the right section without rewriting or corrupting the whole document each turn, and emit those edits in a parseable form your code can apply.

Real-world examples: Claude maintaining a specification document updated with each exchange. Cursor maintaining a codebase summary during a development session. A writing assistant building an article

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section by section.

🛠️ Tools

[T1] MCP (⭐⭐⭐)

The model decides mid-generation to call an external function instead of answering: it emits a structured request (tool name + arguments), your code executes it, returns the result, and the model continues with it.

Real-world examples: Claude deciding to use its calculator when you ask a math question. A booking agent calling a calendar API to check availability.



This will be tested with both your own and a preexisting MCP tool.

[T2] Sandboxed Code Execution (⭐⭐⭐)

Give the agent a sandboxed runtime (Docker or isolated subprocess) where it runs code it generated, captures stdout and tracebacks, and feeds them back to fix its own errors.

Real-world examples: ChatGPT Code Interpreter executing Python to analyze a CSV. Devin generating, executing, and correcting code in an autonomous loop.

[T3] Web Search (⭐⭐)

Let the agent run a live web query, parse the results, and inject them into context.

Real-world examples: Perplexity AI citing web sources in real time. ChatGPT Search. A monitoring agent tracking industry news and producing daily summaries.

[T4] File Access within Permissions (⭐)

Filesystem operations (read, write, create...) + A permission layer defining which files and directories the agent may read or touch, checked before any filesystem operation.

Real-world examples: A code review agent authorized to read and write in `/src`. Claude Cowork requesting explicit permissions before accessing system folders.

[T5] Structured Output (⭐)

Force the model to output valid JSON conforming to a defined schema, instead of free-form natural language prose.

Real-world examples: OpenAI's Structured Outputs API. Any pipeline that parses LLM responses to feed a database or interface.

📚 RAG

[R1] Basic RAG (⭐⭐)

Full pipeline: ingest local documents, chunk them, generate embeddings, store in a local vector store, retrieve by semantic similarity, and inject the top chunks into context. The canonical way to give an agent knowledge it was not trained on.

Real-world examples: A legal assistant answering from a contract database. Google's NotebookLM enabling conversation with uploaded documents. A support chatbot drawing from product documentation.

[R2] Advanced RAG (⭐⭐⭐)

Extends R1 with reranking of retrieved chunks, metadata filtering, and per-chunk relevance scoring to drop near-misses before they reach the context.

Real-world examples: An enterprise document search system filtering by date, author, and department before answering. AI search engines combining vector retrieval and semantic reranking.

🤖 Orchestration & Reasoning

[A1] Reasoning Loops (ReAct) (⭐⭐⭐)

A family of loops that make the model's thinking explicit and iterative. Central to agentic understanding.

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Real-world examples: Claude's o1/o3 ”thinking” mode. Code generation systems that verify their own output before proposing it. Anthropic's Constitutional AI uses a similar principle. Any context where reasoning traceability is as important as the result (medical, legal, financial).

[A2] Multi-Agent Orchestration (⭐⭐⭐)

Multi-agent orchestration has several specialized agents coordinate on a task, passing work between them.

Real-world examples: A support system routing technical questions to an expert agent and billing questions to a commercial agent. Production LLM Routers sending simple requests to lightweight models and complex requests to powerful ones to optimize costs. Microsoft's AutoGen where agents debate to refine a response. CrewAI where a researcher agent, a writer agent, and a critic agent collaborate on an article.

[A3] Human-in-the-Loop (⭐)

Before executing a critical or irreversible action, the agent suspends, presents its plan to the user, and only proceeds after explicit confirmation.

Real-world examples: Claude Cowork asking for confirmation before modifying system files. Automated trading agents presenting their strategy before execution. Any automation system where an irreversible action is at stake.

[A4] Prompt Versioning (⭐)

Treat prompts as versioned artifacts: each version timestamped, comparable, and rollback-able. Run controlled comparisons and measure differences in quality, latency, or behavior.

Real-world examples: Every team maintaining production prompts uses some form of versioning. PromptLayer, Langfuse, and other LLM observability tools integrate this natively.

[A5] Persona (⭐)

The agent's name, tone, role, and response scope are configurable without modifying the code. A configuration file is enough to turn the same system into a legal assistant or a mathematics tutor.
The system manages multiple distinct agent profiles, dynamically selectable and switchable at runtime **without clearing the conversation history** — the agent changes persona mid-session while retaining full context.

Real-world examples: A support system that routes to a technical agent or a commercial agent based on the client profile. An educational platform with one agent per subject.

[A6] Semantic Cache (⭐⭐)

Cache responses and serve them directly for semantically similar queries (matched via embeddings), skipping the model entirely. Cuts latency and CPU load on repetitive requests.

Real-world examples: GPTCache, Redis AI's semantic cache. A support assistant that instantly answers frequent questions without querying the model each time.

[A7] Automated Prompt Optimization (⭐)

Build prompts dynamically from context: auto-selected few-shot examples, adaptive reformulation, or iterative optimization against a quality signal.

Real-world examples: Stanford's DSPy which automatically optimizes prompt pipelines. Systems that dynamically select the best few-shot examples based on the incoming request.

🧪 Evaluation

[EV1] Automated Eval & Hallucination Detection (⭐⭐⭐)

A system automatically evaluates the quality of the agent's responses: relevance, coherence, source faithfulness. An LLM acts as judge and produces usable scores for comparing system versions. Unverifiable or source-contradicted statements are identified and flagged.

Real-world examples: RAGAS for evaluating RAG pipelines. The internal benchmarks all AI teams use to validate their changes before deployment. Princeton's FActScore evaluating response factuality. RAG systems comparing responses to retrieved sources to detect discrepancies.

[EV2] Content & Privacy Guardrails (⭐⭐⭐)

Interceptors on input and output. On input: detect and block unsafe or out-of-scope prompts before they reach the model. On output: detect PII or other inapropriate content, and anonymize or refuse, per configurable rules.

Real-world examples: OpenAI Moderation API content filters. Nvidia NeMo Guardrails. An HR assistant that refuses to answer questions about individual colleagues' salaries. A medical assistant anonymizing patient data before sending it to the model.

[EV3] Logging & Monitoring Dashboard (⭐⭐)

Structured logging of every interaction — prompt sent, response, tools called, latency, token counts — feeding a simple metrics view (call volume, latency, error rate, drift over time).

Real-world examples: Langfuse, LangSmith, Helicone — all LLM observability tools are built on this principle. A dashboard that reveals a prompt modified yesterday has doubled average latency.

[EV4] Adversarial Testing (⭐⭐)

A test suite that attacks your own agent: jailbreak attempts, prompt injections, malformed inputs. Documents where it breaks and where it holds.

Real-world examples: Anthropic, OpenAI, and Google red teams testing their models before deployment. Public adversarial benchmarks (AdvBench, HarmBench).

💬 UX & Lifecycle

[X1] Web UI (⭐)

A decoupled web frontend over the CLI backend: message input, streaming conversation display, live token/context usage, basic agent switching, configuration... The backend logic stays unchanged.

[X2] Branching — Resume & Interrupt (⭐⭐)

Resume: edit a past message and fork an alternative branch, preserving the original history.
Interrupt: abort a generation mid-stream and redirect it with new instructions without waiting for it to finish.

Real-world examples: ChatGPT's and Claude's ”Edit message” button generating a new branch. Prompt prototyping tools that allow exploring multiple directions from the same point. A long-form writing agent whose direction can be corrected mid-stream without waiting for completion.

[X3] Export & Publication (⭐)

Export conversation history and artifacts to usable formats (JSON, Markdown, PDF), and generate a public, no-auth link to view a shared conversation or artifact in a browser.

Real-world examples: ChatGPT's and Claude's ”Share conversation” button generating a public link. PDF export of a working session for archiving or transmission.

[X4] Questions & Forms (⭐)

Add a way for users to answer questions asked by your LLM in a seemless way

[X5] Scheduling (⭐)

The agent can be set up to automatically execute tasks at defined intervals, without manual intervention. Daily summaries, weekly reports, continuous monitoring.

Real-world examples: A monitoring agent sending a news summary every morning. A system monitoring agent checking status every hour and alerting on anomalies.


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